size_balanced_max_legacy

equity strategy | family: equity-max | status: reference | derived from `size_balanced_max`

What this strategy does

size_balanced with bigger bets for more return: base unit 0.009, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. PEAD is stronger in small caps; balancing by size keeps the edge but removes a size bet.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.009
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	size balanced max legacy	market (CRSP VW)
Annualized return	8.45%	9.94%
Annualized volatility	12.26%	19.35%
Sharpe (NAV-pct)	0.72	0.59
Sortino	1.01	
Max drawdown	-37.5%	-55.5%
Calmar	0.23	
Final NAV ($10M start)	$48,527,312	
Worst day	-7.65%	
% positive days	51.2%	

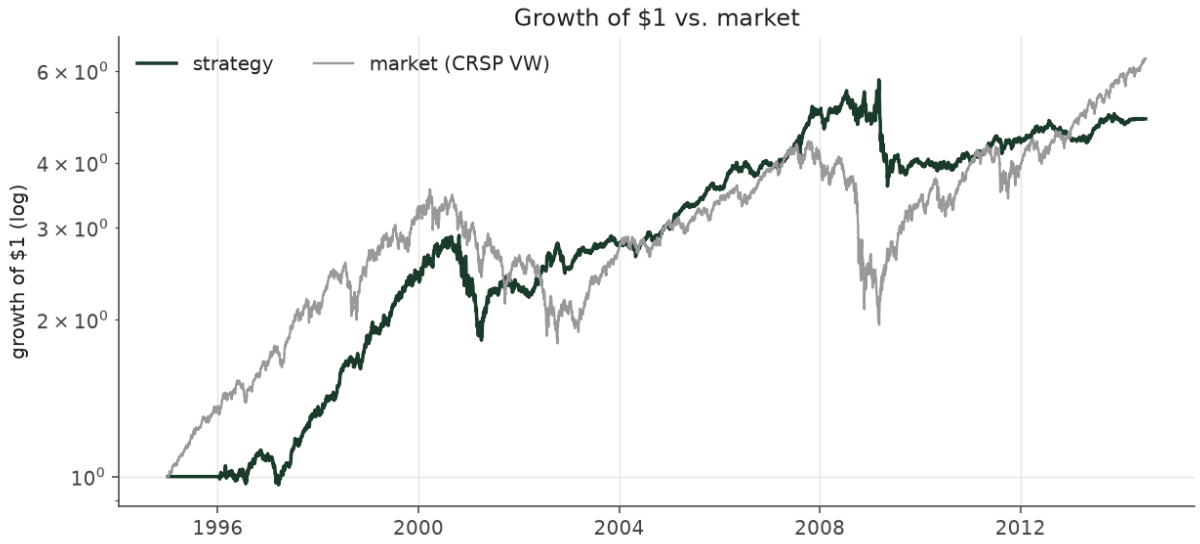
Sharpe 1996-2001	1.00	
Sharpe 2002-2007	1.52	
Sharpe 2008-2013	0.03	

Correlation of daily returns with the market: **0.00**.

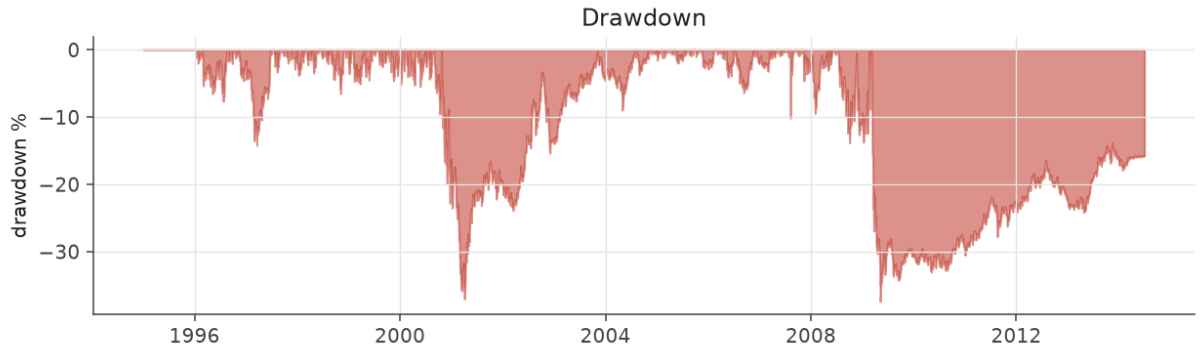
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.723

• best return among equity: `market_boost_max` (0.116); this one: 0.0845

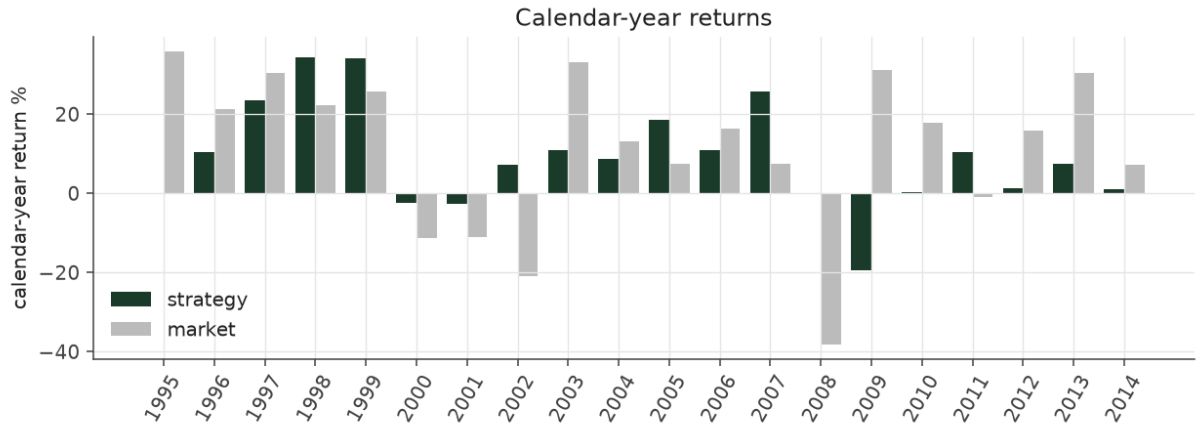
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.226



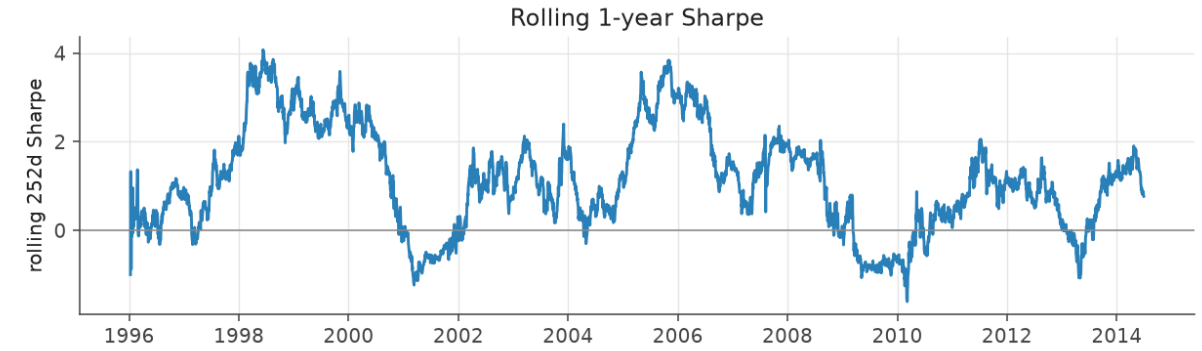
NAV (log scale)



Drawdown from running peak

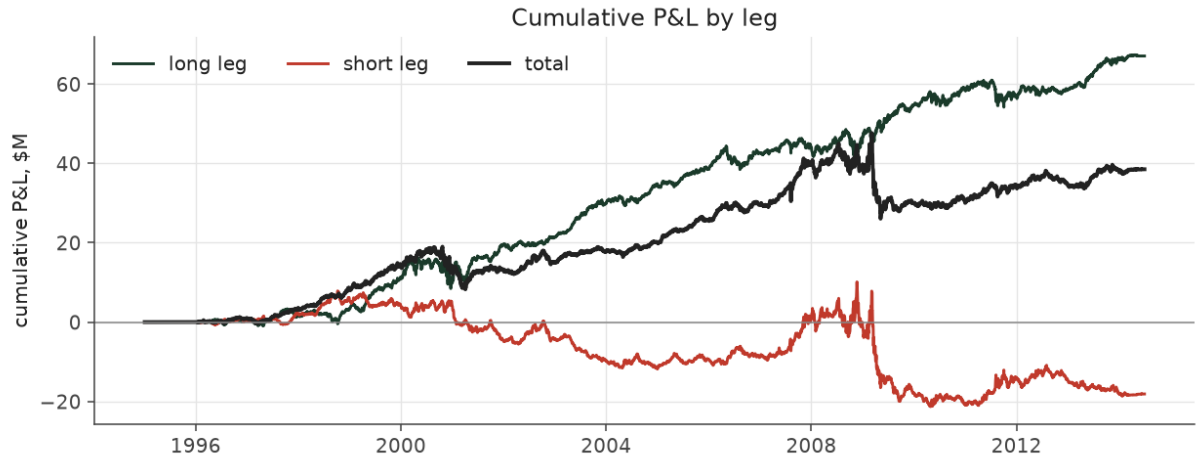


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	67.08	-18.11
P&L after costs ($M)	62.06	-23.53
of which common (average-event) component ($M)	49.67	-47.83

of which excess / PEAD component ($M)	17.28	29.76
excess minus costs ($M)	12.27	24.35
gross return per $ traded (bp)	215.17	-56.29
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	0.95	-0.15

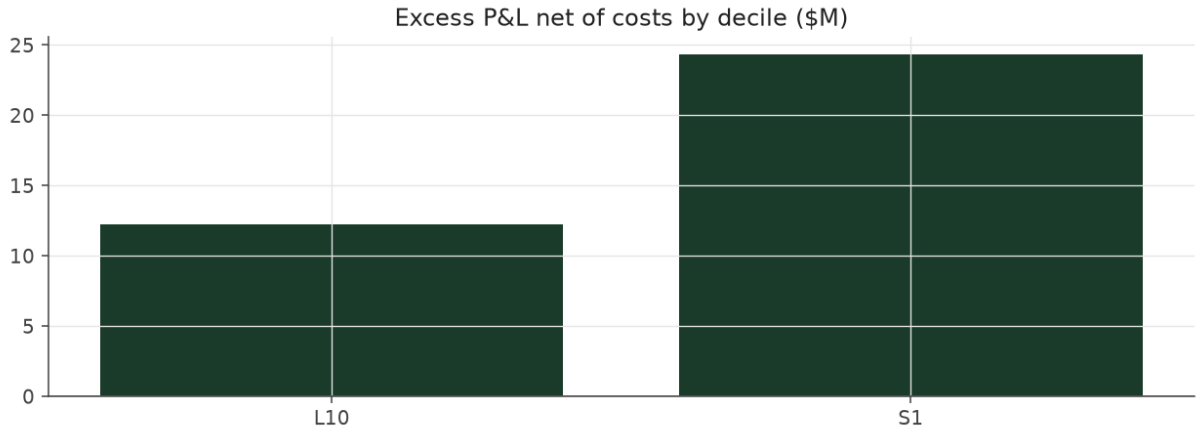
Daily P&L correlation between the two legs: **-0.54**.

• **The short leg loses money in absolute terms** (-18.1M before costs) while the long leg earns +67.1M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 49.7M / 47.8M. Stripping that common component, the surprise signal itself earned +17.3M on the long side and +29.8M on the short side.

• The two legs' daily P&L are -0.54 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.95 long / -0.15 short vs 0.72 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	17,174	3,117.40	62.06	17.28	5.02	12.26	33.5%	39.34
short	1	17,169	3,218.32	-23.53	29.76	5.42	24.34	66.5%	75.63



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,832	217.96	1.03	2.78	0.70	2.07	5.7%	95.08
2	7,159	851.52	9.00	11.39	2.04	9.36	25.5%	109.89
3	7,735	1,508.31	19.31	20.15	2.85	17.31	47.3%	114.75
4	7,731	1,825.97	4.75	6.27	2.73	3.54	9.7%	19.40
5	6,886	1,931.95	4.43	6.45	2.12	4.32	11.8%	22.38

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $

1	4,357	95.80	2.67	3.36	0.48	2.88	7.9%	300.67
2	7,223	626.01	4.82	7.35	1.88	5.47	14.9%	87.38
3	7,834	1,653.81	21.64	24.45	3.31	21.14	57.7%	127.83
4	7,850	2,025.08	3.79	6.43	2.83	3.59	9.8%	17.73
5	7,077	1,934.54	5.72	5.56	1.94	3.63	9.9%	18.75

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	6,338	1,638.57	-1.43	14.57	2.33	12.24	33.4%	74.71
41-60d	11,202	1,959.72	6.23	6.44	3.22	3.22	8.8%	16.44
61-90d	12,665	2,326.18	15.79	8.60	3.90	4.70	12.8%	20.19
>90d	4,138	411.25	17.94	17.43	0.98	16.44	44.9%	399.83

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	9,920	1,114.42	12.42	13.36	1.76	11.60	31.7%	104.09
2002-07	11,506	2,156.30	24.73	27.02	3.52	23.49	64.1%	108.95
2008-13	12,917	3,064.99	1.38	6.66	5.15	1.51	4.1%	4.93

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	6,718	1,160.09	9.46	5.21	1.88	3.33	9.1%	28.71
Chems	687	139.18	0.43	0.54	0.22	0.32	0.9%	22.63
Durbl	869	140.56	-0.42	0.35	0.25	0.10	0.3%	7.37
Enrgy	1,375	271.72	5.57	6.24	0.44	5.80	15.8%	213.39
Hlth	3,037	511.47	3.67	1.92	0.87	1.05	2.9%	20.54
Manuf	3,627	669.11	1.81	3.76	1.08	2.67	7.3%	39.96
Money	5,366	1,096.02	1.04	9.58	1.78	7.80	21.3%	71.18
NoDur	1,540	295.77	2.35	2.46	0.48	1.98	5.4%	66.94
Other	5,074	881.82	6.37	6.73	1.57	5.16	14.1%	58.52
Shops	3,547	641.11	6.64	7.04	1.05	5.99	16.3%	93.41
Telcm	1,164	224.12	1.16	2.13	0.35	1.79	4.9%	79.76
Utils	1,339	304.75	0.44	1.08	0.47	0.61	1.7%	20.12

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	11,517	2,311.71	3.82	9.45	3.51	5.94	16.2%	25.68

low vol	11,554	1,942.27	17.45	19.82	3.47	16.35	44.6%	84.16
mid vol	10,730	1,995.54	16.79	17.50	3.30	14.21	38.8%	71.19

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.45%	0.00%
0.5%	3.4	8.05%	0.40%
1.0%	6.8	7.62%	0.83%
2.0%	13.5	6.64%	1.80%
5.0%	33.8	2.00%	6.44%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.